

WLM Foundation Series

Paper 3

Recursion–Structure Series

The Uniqueness and Unskippability of Recursion: From $\pm\epsilon$ Alternation to the Unique Generative Engine

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Abstract

Paper 1 established the forced transition $0 \rightarrow \epsilon$: the state of perfect symmetry (0) is unique, the minimal deviation from it (ϵ) is the unique first distinguishable state, and no generative system can bypass this first step. Paper 2 established $\epsilon \rightarrow \{\pm\epsilon, -\epsilon\}$: once ϵ exists, the laws of orientation (Axiom D0) force a unique dual pair of opposite directions — no third independent direction is possible, and duality cannot be bypassed.

This paper establishes the third forced transition: $\{\pm\epsilon, -\epsilon\} \rightarrow$ recursion. Duality alone is not enough to generate structure: a system that can only alternate between two states finitely many times will eventually stagnate. What is needed is the ability to repeat this alternation indefinitely — that is, recursion. Under Axioms R1 (composition closure) and R2 (orientation preservation under iteration), any non-recursive system collapses to trivial dynamics. Recursion is the unique generative engine capable of producing non-trivial invariant structure, and it cannot be bypassed.

Epistemic status. All theorems are provable within the WLM framework using only axioms and results from Papers 0–2. No new axioms are introduced. This paper adds two new partial results toward Conjecture U1 (the conjecture that the full chain $0 \rightarrow \epsilon \rightarrow \{\pm\epsilon\} \rightarrow$ recursion $\rightarrow$ structure $\rightarrow$ appearance is the unique generative chain).

$0 \rightarrow \epsilon$ (Paper 1) $\rightarrow \pm\epsilon$ (Paper 2) $\rightarrow$ recursion (Paper 3) $\rightarrow$ structure (Paper 4) $\rightarrow$ appearance (Paper 5)

0. Framework and Inherited Results

This section collects the axioms and prior results that Paper 3 uses. All of these were established in Papers 0–2; none are new here. Readers unfamiliar with the earlier papers may treat them as the starting conditions of this paper.

The three axioms used in this paper govern how generative rules compose and how they preserve orientation:

Axiom D0 (Generative Orientation and Reversal, from Paper 2). (a) *Primitive Orientation:* Any state x that is distinguishable from 0 carries an orientation label $\theta(x)$, preserved under observational equivalence. (b) *Orientation Reversal:* There exists at least one generator F that flips orientation: $F(+\varepsilon) \sim_o -\varepsilon$ and $F(-\varepsilon) \sim_o +\varepsilon$.

Axiom R1 (Composability). If G and H are generative rules, then $H \circ G$ is also a generative rule. Equivalently: iterates f^n are defined for all $n \geq 1$. This is what makes recursion possible: the ability to apply a rule to its own output, indefinitely.

Axiom R2 (Recursive Orientation Preservation). If a generator f realizes an orientation flip ($f(+\varepsilon) \sim_o -\varepsilon$), then all iterates f^n preserve this alternation pattern. No recursion can ‘wash out’ the directional structure established by duality.

The four key inherited results used without re-proof:

- Paper 0, Lemma 4c: In any non-collapsing generative chain, every non-constant orbit must alternate between $+\varepsilon$ and $-\varepsilon$ indefinitely. This means persistent alternation is not optional — it is the only form a non-trivial orbit can take.
- Paper 0, Theorem 5: Any recursive orbit that neither collapses nor diverges must generate at least one invariant structure — either a periodic cycle or a compact closure (by the Bolzano-Weierstrass theorem).
- Paper 2, Theorems 4.2 and 4.3: The dual pair $\{+\varepsilon, -\varepsilon\}$ is the unique generator of non-trivial recursion, and it cannot be bypassed by any generative chain that reaches non-trivial dynamics.
- Paper 1, Theorem 3.3 and valuation $v(\cdot)$: ε is the unique minimal offset and cannot be skipped; the valuation $v(\cdot)$ measures structural distance from 0.

Remark (How to Read This Paper). Paper 3 follows the same four-section architecture as Papers 1 and 2. Section 1 characterizes the minimal structure of recursion as a generative layer. Section 2 shows that duality forces recursion — without recursion, the system built on $\pm\varepsilon$ collapses. Section 3 classifies all the ways a chain can fail to establish recursion. Section 4 shows that recursion generates structure uniquely and proves it cannot be skipped. The structural argument is analogous to Paper 2’s treatment of duality, which was analogous to Paper 1’s treatment of ε . The parallelism is intentional.

1. Recursion Space

The previous papers established what it means for a generative system to have its first distinction (ε) and its first directionality ($\pm\varepsilon$). This section characterizes what recursion adds: the ability to iterate generative steps indefinitely, rather than stopping after a finite number of applications.

Definition 1.1 (Recursive Generator). A generator $f \in \mathcal{G}$ is recursive if its iterates f^n are defined for all $n \geq 1$ via Axiom R1. The orbit of a state x under f is the infinite sequence $x, f(x), f^2(x), \dots$. A non-recursive generator can be applied only finitely many times before the composition is no longer defined.

1.1 Minimal Recursion

The first result establishes that recursion — not any finite composition — is the minimal engine capable of sustaining what duality requires.

Theorem 1.1 (Minimal Recursion). Under R1 and R2, the minimal non-trivial generative engine is a recursive generator sustaining persistent $\pm\varepsilon$ alternation.

Proof. R1 provides composition closure: any generator can be iterated. A single non-recursive step on $\pm\varepsilon$ yields only a finite sequence. By Paper 0 Lemma 4c, any non-collapsing orbit must alternate $\pm\varepsilon$ persistently — for all n , not just finitely many. R2 ensures that a generator realizing a flip preserves this pattern under all iterates. Therefore the smallest engine exceeding the duality layer is a recursive (infinitely iterable) generator whose iterates sustain persistent $\pm\varepsilon$ alternation. No finite engine suffices. $\square$

1.2 Recursion Irreducibility

The second result shows that recursion cannot be replaced by something simpler: any simpler mechanism either stagnates within $\pm\varepsilon$ or returns to the symmetry interval I_0 .

Theorem 1.2 (Recursion Irreducibility). No non-recursive rule can produce non-constant orbits that escape the $\pm\varepsilon$ layer into higher structure.

Proof. Without R1, only finite compositions exist. A finite chain of generators acting on $\pm\varepsilon$ either (i) reaches a fixed point (the system stagnates), (ii) returns to I_0 (symmetry collapse), or (iii) reaches a state with valuation $v > v(\varepsilon)$, which by Paper 1 Theorem 4.2 requires ε to have appeared earlier — giving no genuinely new structure. In all cases, persistent alternation required by Lemma 4c cannot be sustained without infinite iteration. No non-recursive rule escapes the $\pm\varepsilon$ layer. $\square$

Lemma 1.3 (Recursion as Engine). Recursion is the first mechanism whose orbits can exceed the capacity of any finite number of single generative steps.

Proof. Only recursive iteration can sustain persistent alternation indefinitely (Lemma 4c). By Paper 0 Theorem 5, such sustained non-collapsing recursion generates at least one invariant structure. A finite composition of generative steps produces at most finitely many distinct states, which cannot constitute a persistent alternation for all n . Therefore no finite composition can achieve what recursion achieves. $\square$

2. Recursion Forced by Duality

Paper 2 showed that ε forces duality: once you have a first distinction, you must have two opposite directions. This section shows that duality forces recursion: once you have two opposite directions, you must have an engine that can alternate between them indefinitely. Finite alternation is not enough.

2.1 Duality Requires Recursion

Theorem 2.1 (Duality Requires Recursion). *Given a realized dual pair $\{+\varepsilon, -\varepsilon\}$, any non-recursive system collapses to trivial dynamics.*

Proof. Paper 2 establishes that $\pm\varepsilon$ alternation is forced once duality is realized. However, without R1, only finitely many compositions are possible. A finite alternation must eventually terminate: the system reaches a state where further application either fixes the current state (fixed point) or returns to I_0 (symmetry collapse). Both outcomes contradict Lemma 4c for non-collapsing chains, which requires persistent alternation for all n . Therefore duality is necessary but not sufficient for non-trivial dynamics: recursion (R1) is required to sustain the alternation indefinitely. $\square$

2.2 Recursion Uniqueness

It is not enough to know that recursion is required — we also need to know that no other mechanism could serve the same role.

Theorem 2.2 (Recursion Uniqueness). *Closed composition under R1 is the unique generative engine compatible with persistent $\pm\varepsilon$ alternation.*

Proof. Suppose a non-recursive engine Σ could generate non-trivial orbits from $\pm\varepsilon$ without relying on infinite composition. Such orbits would be non-constant and, by Lemma 4c, would have to alternate $\pm\varepsilon$ persistently for all n . But persistent alternation for all n requires that every iterate f^n is defined — which is precisely what R1 provides. Without R1, iterates are undefined for sufficiently large n , making persistent alternation impossible. Therefore Σ would implicitly require R1 to sustain its orbits, contradicting the assumption that it is non-recursive. No non-recursive engine generates persistent $\pm\varepsilon$ alternation. Recursion (R1) is the unique engine. $\square$

3. Collapse of Non-Recursive Chains

The previous section showed that recursion is required. This section completes the picture by classifying every possible way a generative chain can fail to establish recursion. There are exactly two structural failure modes, plus two derivative modes inherited from earlier layers. Every non-recursive chain falls into one of these categories; there is no other outcome.

3.1 Static Collapse

The first failure mode occurs when a generative system has duality ($\pm\epsilon$) but no closure: each generator can be applied, but the result of applying two generators in sequence is not itself a valid generator. Without this closure, the system can only make finitely many moves before stopping.

Theorem 3.1 (Static Collapse). *If recursion is skipped (R1 not available), the system remains dynamically trivial.*

Proof. Without R1, only finite compositions of generators acting on $\pm\epsilon$ are available. By Theorem 1.2, such finite chains either produce fixed points or return to I_0 . No invariant structure of higher order can form: a finite sequence of states cannot be a periodic orbit (which requires the orbit to close under infinite iteration) or a compact closure (which requires Bolzano-Weierstrass applied to an infinite sequence). The system is statically trapped. $\square$

3.2 Divergent Collapse

The second failure mode occurs when a system attempts to iterate without the closure guarantee of R1: each step is defined, but compositions may escape the structured part of the state space entirely.

Theorem 3.2 (Divergent Collapse). *If recursion is replaced by non-closed iteration (compositions defined only for finitely many steps, or escaping bounded valuation regions), the system diverges without forming invariants.*

Proof. Without R1 closure, repeated application of a generator can produce states of arbitrarily high valuation with no upper bound. By Paper 0 Theorem 5, invariant structure requires a bounded non-collapsing orbit (to which Bolzano-Weierstrass applies). An unbounded orbit has no limit point in any bounded region and thus produces no compact invariant set. The system diverges without generating any stable structure. $\square$

Figure 1: All collapse modes of the recursion layer. Every non-recursive chain falls into one of these four categories.

Collapse Mode	Trigger	Consequence	Reference
Static Collapse	R1 skipped	Fixed points or return to I_0	Thm 3.1
Divergent Collapse	Non-closed iteration	No invariants; unbounded orbits	Thm 3.2
Alternation Failure	R2 violated	Finite flip $\rightarrow$ collapse	P0 Lemma 4c
Duality Without Engine	$\pm\epsilon$ present, no recursion	Trivial dynamics	Thm 2.1

4. Recursion Generates Structure Uniquely

The previous sections established what recursion is (Section 1), why it is forced (Section 2), and what happens without it (Section 3). This section establishes the positive claim: recursion not only is necessary, but is sufficient for generating the first non-trivial invariant structure. And crucially, no generative chain that reaches invariant structure can have bypassed the recursion layer.

4.1 Invariant Generation

Theorem 4.1 (Invariant Generation). *Under R1 and R2, recursion generates the minimal non-trivial invariant sets.*

Proof. By Paper 0 Theorem 5, any recursive orbit that neither collapses nor diverges is either eventually periodic (generating an invariant cycle: a finite set of states that the orbit visits repeatedly forever) or infinite and bounded (generating a compact invariant closure by Bolzano-Weierstrass: a limit set that the orbit approaches). Combined with R2-persistent $\pm\epsilon$ alternation (Lemma 4c), recursion produces the first stable structures beyond the duality layer — structures that are neither trivial fixed points nor mere alternations, but genuine invariant sets with non-trivial geometry. $\square$

4.2 No Structure Without Recursion

Theorem 4.2 (No Structure Without Recursion). *Any attempt to reach invariant structure without recursion collapses.*

Proof. For a set S to be invariant under a generative rule f , we need $f(S) \subseteq S$: the rule maps S into itself. This is a closure condition. For S to be non-trivially invariant under persistent $\pm\epsilon$ alternation, the closure must hold under infinite iteration of f — which is exactly what R1 provides. Without R1, no such closed set exists: by Theorems 3.1 and 3.2, the system either stagnates (no set large enough to be non-trivially invariant forms) or diverges (no bounded set is preserved). Therefore invariant structure requires recursion. $\square$

4.3 Recursion is Unskippable

This is the main result of the paper, and the direct analogue of Paper 1's ' ϵ cannot be skipped' and Paper 2's ' $\pm\epsilon$ cannot be skipped.'

Theorem 4.3 (Recursion is Unskippable). *Any generative chain realizing non-trivial invariant structure must pass through the recursion layer. No chain generates invariant structure while bypassing recursion.*

Proof. Contrapositive: suppose a chain bypasses recursion. Then either R1 is absent (the chain undergoes static collapse by Theorem 3.1: no invariants form) or R1 is present but not applied to persistent alternation (the chain undergoes the failure described in Theorem 4.2: no invariant structure is reachable). In both cases of recursion absence, non-trivial invariant structure is impossible. Therefore any chain that does generate non-trivial invariant structure must have passed through the recursion layer. This is the recursion

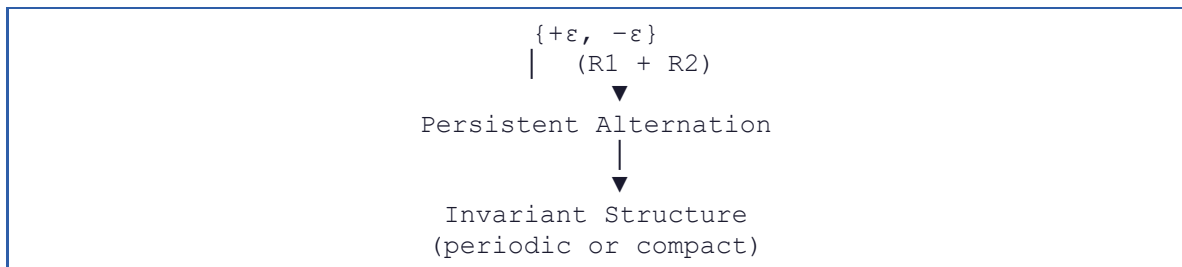
analogue of Paper 1 Theorem 3.3 (ε unskippable) and Paper 2 Theorem 4.3 ($\pm\varepsilon$ unskippable), and it constitutes a new partial result for Conjecture U1. $\square$

Remark (Unskippability Chain). *The three unskippability results established so far are: ε cannot be skipped (Paper 1 Thm 3.3): any chain reaching non-zero states must pass through ε . $\pm\varepsilon$ cannot be skipped (Paper 2 Thm 4.3): any chain generating non-trivial recursion must pass through duality. Recursion cannot be skipped (Theorem 4.3 of this paper): any chain generating non-trivial invariant structure must pass through recursion. Each result adds one confirmed link toward Conjecture U1, which asserts that the full chain is the unique forced sequence with no alternative paths.*

5. Structural Summary

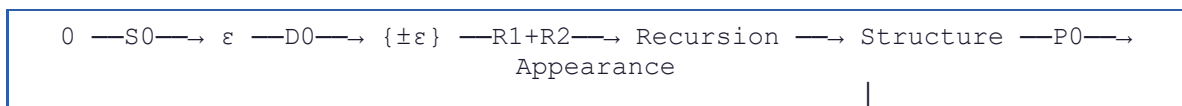
R1 closes composition, making infinite iteration defined. R2 preserves directional alternation under iteration. Together they transform finite $\pm\varepsilon$ alternation into persistent recursive dynamics, which Paper 0 Theorem 5 guarantees will generate invariant structure. Without either axiom, only static or divergent collapse is possible. Recursion is the unique engine and cannot be skipped.

Figure 1: The Recursion Layer



The recursion layer takes the dual pair $\{+\varepsilon, -\varepsilon\}$ as input and produces invariant structure as output. The transition requires both R1 (so that alternation can be iterated indefinitely) and R2 (so that the directional pattern of alternation is preserved under iteration). Neither axiom alone suffices.

Figure 2: Full Generative Chain



The full chain from Papers 0–3. Each arrow is labeled with the axiom or prior result that forces the transition. Collapse paths branch off the recursion layer: any failure to realize recursion sends the chain into one of the modes classified in Section 3.

Figure 3: Parallelism Across Papers 1–3

Layer	Paper	Forced Step	Key Property	Unskippable Result
Offset	1	$0 \rightarrow \epsilon$	Minimal, unique	Thm 3.3
Duality	2	$\epsilon \rightarrow \{\pm\epsilon\}$	Binary orientation	Thm 4.3
Recursion	3	$\{\pm\epsilon\} \rightarrow \text{Recursion}$	Unique engine	Thm 4.3

Each paper establishes one layer of the generative chain. The structure is identical: prove minimality and uniqueness of the new layer, show it is forced by the previous layer, classify all collapse modes, prove unskippability. The three unskippability theorems together form the partial results for Conjecture U1.

Conclusion

Paper 3 establishes the third forced link of the WLM generative chain: $\{+\epsilon, -\epsilon\} \rightarrow \text{recursion}$. The argument proceeds in four steps, mirroring the structure of Papers 1 and 2. First, recursion is characterized as the minimal engine capable of sustaining persistent $\pm\epsilon$ alternation (Section 1). Second, duality is shown to require recursion: without R1, $\pm\epsilon$ alternation cannot be sustained and the system collapses (Section 2). Third, all ways in which a chain can fail to establish recursion are classified: static collapse and divergent collapse, plus two inherited failure modes (Section 3). Fourth, recursion is shown to generate invariant structure uniquely and to be unskippable: no chain reaches non-trivial structure without passing through recursion (Section 4).

The three unskippability results established in this series — ϵ (Paper 1), $\pm\epsilon$ (Paper 2), recursion (Paper 3) — are partial results toward Conjecture U1, which asserts that the full chain $0 \rightarrow \epsilon \rightarrow \{\pm\epsilon\} \rightarrow \text{recursion} \rightarrow \text{structure} \rightarrow \text{appearance}$ is the unique forced sequence. Paper 4 will establish the fourth forced transition: recursion $\rightarrow$ invariant structure.

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Bolzano-Weierstrass theorem. — Used in Paper 0 Theorem 5, inherited here: any bounded infinite sequence has a convergent subsequence, providing the compact invariant closures that recursion generates.

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